

Project Report 1

SEP 2026 SEMESTER

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BACHELOR OF COMPUTER SCIENCE

MODELLING AND SIMULATION

TEB2103

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1. Introduction

In our project background, we proposed a nanobot swarm that follows chemical signals to help a skin wound heal faster. For this assignment, we built an agent-based model (ABM) in Python using Mesa [1] to compare the healing time of a signal-following swarm, a random swarm and no nanobots. The code is at https://github.com/Nowkee/M8S_Assignment.

2. The Model

The skin is a 40 by 40 grid with an oval wound of 331 squares, which starts with four random infection patches (about 90 squares) that grow and spread. Unhealed squares give off a chemical signal that spreads and fades, while the body slowly heals from the edges but never an infected square.

We placed 100 nanobots at the wound edge, made up of 20 scouts that mark unhealed squares, 40 cleaners that kill bacteria and 40 carriers that add growth factor to clean squares. They follow the five rules from our proposal, moving towards the strongest signal (randomly 15% of the time) with limited fuel and medicine. The random swarm is identical, but always moves randomly.

Healing time is the number of steps until 95% of the wound is healthy. Each case was run 20 times for 400 steps on the same 20 random wounds.

3. Results

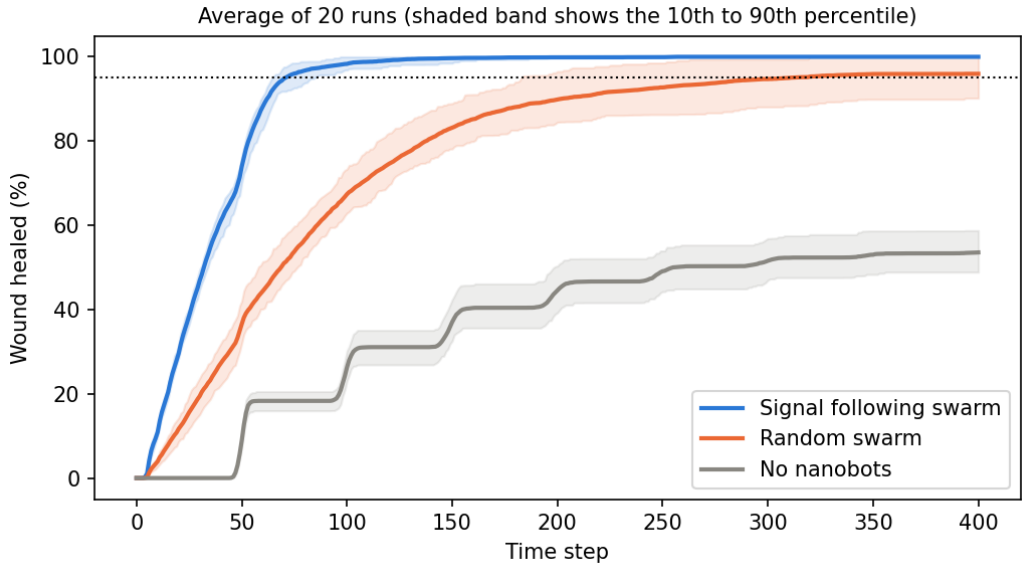


Figure 1 shows the average percentage of the wound healed over time, with the dotted line as the 95% finish line.

Case	Runs that reached 95% (out of 20)	Mean healing time (steps)	Healed after 400 steps	Infected squares after 400 steps
Signal-following swarm	20	72.5 (sd 8.5)	100%	0

Case	Runs that reached 95% (out of 20)	Mean healing time (steps)	Healed after 400 steps	Infected squares after 400 steps
Random swarm	13	242.2 (sd 45.7)	96.0%	13.3
No nanobots	0	Did not heal	53.6%	153.4

Table 1 shows the results from 20 runs of each case. The random swarm time only averages its 13 finished runs.

The signal-following swarm healed in every run and was about 3.3 times faster than the random swarm, whose bots drift into healthy skin and waste their fuel. Without nanobots, healing got stuck at around 54% as the infection grew to about 153 squares.

4. Discussion and Conclusion

Since both swarms had the same bots, fuel and medicine, the gap comes only from following the signal, with no leader needed [2]. However, the parameters were picked by us, so the step counts are not real time, and the model is flat and ignores blood flow. All in all, the way the swarm moves matters as much as having the nanobots there.

References

[1] Mesa: Agent-based modeling in Python. [Online]. Available: https://mesa.readthedocs.io

[2] R. Wu, Y. Zhu, X. Cai, S. Wu, L. Xu, and T. Yu, "Recent process in microrobots: From propulsion to swarming for biomedical applications," Micromachines, vol. 13, no. 9, Art. no. 1473, 2022, doi: 10.3390/mi13091473.

